

Activity - 1

2a/9/26
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1) $\boxed{54} \quad \boxed{\frac{52}{54}} \xrightarrow{\text{pop}} \boxed{54} \quad \boxed{\frac{55}{54}} \quad \text{push} \quad \boxed{\frac{62}{54}} \quad \text{pop} = 62$

$[21] \quad [21, 24] \xrightarrow{\text{dequeue}} [24] \quad [24, 28] \quad [24, 28, 32] \quad [28, 32]$
dequeue = 24

$S + Q = 62 + 24 = 86$

2) postfix expression

$10 \ 5 \ + \ 60 \ 6 \ / \ * \ 8$

$\boxed{\frac{5}{10}} + \boxed{\frac{6}{60}} \div \boxed{\frac{60}{15}} \quad \boxed{60/6=10} \quad \boxed{\frac{6}{60}} \quad \boxed{\frac{15}{10}} \rightarrow \boxed{\frac{150}{10}} \rightarrow \boxed{15}$
 $15 \times 10 = 150$
 $150 - 8 = 142$

3) postfix expression with single-digit operands

$8 \ 2 \ 3 \ ^1 / 2 \ 3 \ * \ + \ 5 \ 1 \ * \ -$

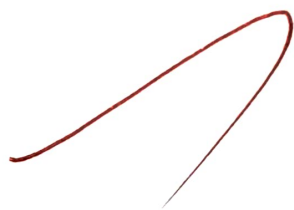
$\boxed{8} \quad \boxed{\frac{2}{8}} \quad \boxed{\frac{3}{2/8}} \xrightarrow{1} \boxed{\frac{8}{8}} \quad \boxed{\frac{1}{8}} \quad \boxed{1} \quad \boxed{\frac{2}{1}} \quad \boxed{\frac{3}{2/1}} \xrightarrow{*} \boxed{\frac{6}{1}} \xrightarrow{+} \boxed{6}$
 $\boxed{7} \quad \boxed{\frac{5}{7}} \quad \boxed{\frac{3}{7}} \quad \boxed{\frac{5}{7}} \quad \boxed{2}$

4) $a + b \times c - d \wedge e \wedge f$

char	stack	o/p
a	a	a
+	+	a
b	+	ab
*	+	abc
c	+	abc
-	-	abc*
d	-	abc*+
^	-	abc*+d
e	-	abc*+d
f	-	abc*+d

e	-	abc*+d
A	-	abc*+d
f	-	abc*+d
-	-	abc*+d

$$\begin{aligned}
 5) \quad & \boxed{10} \rightarrow \boxed{\begin{smallmatrix} 20 \\ 10 \end{smallmatrix}} \xrightarrow{\text{Pop}} \boxed{20} \rightarrow \boxed{\begin{smallmatrix} 10 \\ 10 \end{smallmatrix}} \rightarrow \boxed{\begin{smallmatrix} 20 \\ 10 \\ 10 \end{smallmatrix}} \rightarrow \\
 & \xrightarrow{\text{Pop}} \boxed{\begin{smallmatrix} 10 \\ 10 \end{smallmatrix}} \xrightarrow{\text{Pop}} \boxed{10} \xrightarrow{\text{Pop}} \boxed{} \rightarrow \boxed{20}
 \end{aligned}$$



Amal
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